

# RUPESH SINGH KARKI

 [rupesh](#) |  [rupesh](#) |  [portfolio](#) |  [karkisinghrupesh@gmail.com](mailto:karkisinghrupesh@gmail.com) |  +91 9355-773-350

## WORK EXPERIENCE

### SociolinQ | Full Stack Developer Intern

Aug 2025 – Present

- Architected a **role-based access control (RBAC)** system and **multi-level approval workflows**, enabling secure and structured data operations across organizational hierarchies.
- Designed and implemented **approval-driven data workflows** ensuring that all critical information changes pass through designated approvers, maintaining **data integrity** and enforcing secure modifications in controlled environments.
- Leveraged **Redis** for caching and state management, reducing redundant database queries and improving response times for frequently accessed endpoints.
- Orchestrated serverless cron jobs using **AWS EventBridge** and **AWS Lambda** to automate scheduled background tasks and system maintenance.
- Integrated **AWS CloudWatch** for centralized logging and continuous monitoring of company data, enhancing infrastructure visibility and proactive issue resolution.
- Developed a fault-tolerant **S3 transaction handler** with rollback mechanisms, maintaining data consistency during partial failures in file upload workflows.
- Applied backend optimization techniques such as **request batching**, **caching strategies**, and efficient schema design to improve system scalability and reliability.

## PROJECTS

### Eidos: Self-Healing RAG System

[GitHub](#) [Live](#)

- Developed a production-grade self-healing RAG system using **FastAPI**, **LangGraph**, **Gemini**, **Qdrant**, **PostgreSQL**, **Cloudflare R2** and **NextJs**, automatically evaluating and refining low-confidence responses through a critic-in-the-loop workflow. Containerized with Docker and deployed on **AWS** using **Jenkins CI/CD**, **Elastic IP**, and **Nginx** reverse proxy.

### HearU

[GitHub](#)

- Built an AI-powered mental wellness platform using **React**, **FastAPI**, **PostgreSQL**, and **Gemini**, enabling secure journaling, voice interactions, and personalized guidance. Implemented JWT authentication, REST APIs, and automated CI/CD pipelines with deployment on **Google Cloud** behind **Nginx**.

## PUBLICATIONS

A Comprehensive Survey in ANN-based Customer Churn Prediction, In ICDAM-2025 ([Link](#))

## EDUCATION

|                |                                                        |                                           |
|----------------|--------------------------------------------------------|-------------------------------------------|
| 2023 – present | <b>Maharaja Agrasen Institute of Technology (MAIT)</b> | B.Tech in Computer Science and Technology |
|----------------|--------------------------------------------------------|-------------------------------------------|

## SKILLS

|             |                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------|
| Programming | C/C++, Python, TypeScript, JavaScript, Golang                                                                |
| Libraries   | LangGraph, TensorFlow, Scikit-learn, FastAPI, React.js, Express.js, Chi                                      |
| Databases   | PostgreSQL, MongoDB, Qdrant, Milvus                                                                          |
| Tools       | Docker, Redis, RabbitMQ, Celery, Redux, Git, GitHub Actions, Jenkins, Nginx, AWS EC2, Alembic, Cloudflare R2 |

## CERTIFICATIONS

HackerRank Problem Solving (Basic) ([Certificate](#))